

Smart Metro Ticket Booking System

Project Design Phase

Date	
Team ID	
Project Name	Smart Metro Ticket Booking System
Maximum Marks	4 Marks

2. Proposed Solution

The proposed solution is a ServiceNow-based Smart Metro Ticket Booking system that lets passengers search routes, view fares, pay digitally, and receive a QR ticket in minutes — while giving Station Managers, Metro Operations, and IT Admins the tools to manage stations, monitor performance, and configure the platform without counter dependency or custom code.

2.1 Solution Overview

At its core, the solution digitizes the entire booking-to-travel journey. A passenger selects a source and destination from a searchable catalog form, sees an instant itemized fare (with passenger-type discounts auto-applied), pays through UPI, card, or wallet, and receives a unique, single-use QR ticket delivered by app notification, email, and WhatsApp/SMS. The same QR code is validated at the station gate in real time, and every step is logged for audit, reporting, and exception handling.

2.2 End-to-End Passenger Journey

Proposed Solution — Passenger Booking Journey

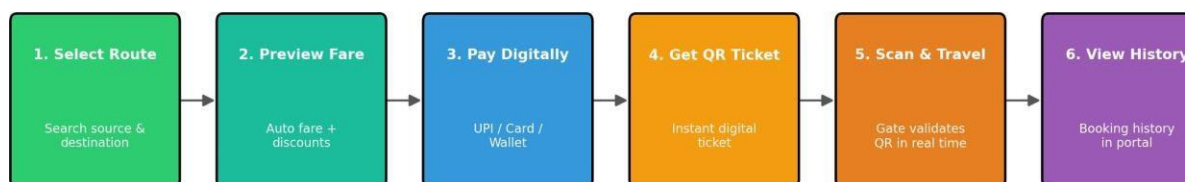


Figure 6.1 — Passenger journey from route selection to travel

2.3 Key Capabilities

- Searchable Service Catalog booking form with source/destination validation against the Station Master.
- Real-time, itemized fare calculation with automatic Student/Senior Citizen discounts.
- Digital payments via UPI, debit/credit card, and mobile wallets with a clear retry path on failure.
- Instant, single-use QR ticket generation linked to the passenger's booking record.
- Multi-channel delivery of tickets and alerts — in-app, email, and WhatsApp/SMS.

- Booking history and ticket re-download through the Service Portal.
- Station Manager tools for master-data maintenance and real-time gate validation.
- Metro Operations dashboards for ticket volume, revenue split, and peak-time analytics.
- IT Admin configuration of catalog variables, fare rules, and role-based access without code changes.

2.4 Value Proposition

Persona	Value Delivered
Passengers	Book and travel in under 5 steps, with no queue or cash needed.
Station Managers	Faster, verifiable gate checks and less manual data upkeep.
Metro Operations	Data-driven scheduling and revenue tracking, updated in near real time.
IT Admins	Faster turnaround on fare and configuration changes, with full traceability.